DETAILED DESCRIPTION OF THE INVENTION

Reconstruction Algorithms and Geometry Profile Generation

Distance measurements obtained from one or more time-of-flight (ToF) sensors and correlated longitudinal displacement data obtained from optical motion sensing components may be processed through a computational reconstruction engine configured to generate a structured three-dimensional representation of penile anatomy. In various embodiments, distance samples are temporally synchronized with positional data to establish a sequence of spatially indexed radial measurements along a defined longitudinal axis.

The reconstruction engine may interpolate intermediate surface points between sampled positions using spline interpolation, polynomial fitting, cylindrical approximation modeling, parametric surface fitting, or mesh generation techniques. Cross-sectional profiles may be constructed at discrete longitudinal intervals, and radial distributions may be calculated for each interval. Curvature analysis may be performed by evaluating changes in centroid alignment across successive cross-sections. Volumetric estimation may be derived through numerical integration of cross-sectional areas along the longitudinal axis.

In certain embodiments, noise filtering and error correction routines may be applied to remove outlier distance values, compensate for minor motion inconsistencies, and smooth discontinuities in surface reconstruction. Data fusion algorithms may integrate measurements from multiple ToF sensors, multi-zone depth arrays, optical flow sensors, and inertial measurement units to produce a corrected spatial model. Weighting functions may prioritize higher-confidence measurements based on signal quality metrics or physiological validation inputs.

The resulting reconstructed model may be converted into a standardized digital geometry profile comprising a structured dataset of derived metrics. Such metrics may include total longitudinal length, cross-sectional radius distribution, circumferential variability, taper gradients, curvature vectors, symmetry indices, proportionality ratios, surface continuity parameters, and volumetric descriptors. The geometry profile may further include normalized indices generated by comparing measured parameters against stored reference distributions.

In some embodiments, the geometry profile generation module may encode the reconstructed data into a compressed parametric representation suitable for secure storage, transmission, or integration with external systems. Profile generation may occur on-device, within a companion application, or within a remote processing environment. The specific mathematical modeling techniques, interpolation strategies, or data structures employed may vary without departing from the core functionality of generating a user-specific penile anatomical geometry profile from synchronized distance and motion measurements.